DefeasibleGalaxies

Project for

Ethics in Artificial Intelligence

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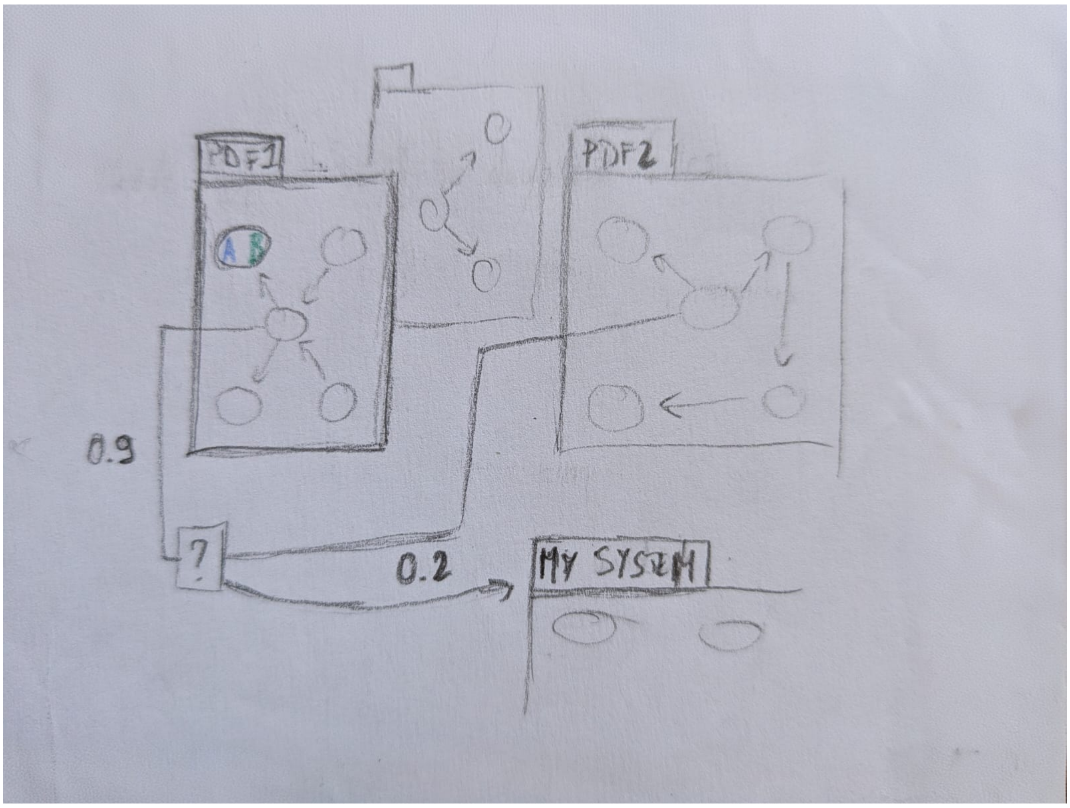

July 15, 2026

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1 Introduction

The 24/7 flood of information makes verifying novel claims increasingly difficult. Beyond our current "infodemic" era, real-time verification is crucial in high-stakes scenarios like live debates.

DefeasibleGalaxies addresses this by leveraging modern NLP and AI to empower machines to reason over claims against a specific knowledge base, providing objective evaluations. The prototype will model each source as a ***system*** of nodes and edges. By cross-referencing how sources support or contradict one another, it maps a comprehensive ***galaxy*** of interconnected information.

1.1 Bridge with the lectures

Argumentation was the most captivating lecture topic. *While my initial idea to integrate this exam-project with my WEFE drone thesis proved impractical, I retain a strong personal interest in diving into this domain, whether or not it ultimately features in my thesis.* This choice is driven by the challenges outlined above, romanticized through the **astronomic** metaphor: individual argumentation frameworks form distinct **systems** (nodes and edges), which interconnect to create expansive **galaxies** of reasoning.

2 Exam goal: description of DefeasibleGalaxies

2.1 Minimum delivery

Translating the complex nuances of natural language into formal logic requires a structured approach. The minimum viable delivery consists of:

1. **Literature Review & Framework Selection:** A comprehensive review will establish the foundation for the software design. The goal is to identify current SOTA solutions in **Argumentation mining** and evaluate if **Dung's abstract framework** is sufficient for our baseline. Furthermore, a **Defeasible logic framework** will be implemented as the Structured Argumentation module to instantiate the abstract model, allowing a single knowledge base to yield different interpretations based on specific premises and consequences.
2. **First milestone — analyzing a single PDF:** After selecting the formalism, a "scrape-understand-build" pipeline will be developed and tested on a single academic PDF (representing a single system). The system will then evaluate a user-provided claim by analyzing it against the generated graph to return a support, rejection, or inconclusive verdict.
3. **Testing:** The extraction pipeline and chosen formalism will be validated against established Argumentation Mining benchmarks.
4. **Human feedback interface:** A basic Web application will visually translate the machine's reasoning for the user. It will display the user's claim alongside the generated argumentation framework, highlighting nodes and edges based on argument weight and intensity.

So, in the end, **the expected deliverable consists of**:

A tool for humans: a **Web app** that receives a PDF (fig.2) and **triggers the pipeline to generate the argumentation framework**, allowing users to visually analyze how the source supports or defeats specific claims (fig.3).

(The Login screen (fig.1) is included for completeness and potential expansion in point 2.3.2)

The interface will loosely resemble a chatbot; however, rather than generating conversational text, each user prompt will generate a static report evaluating the claim's support.

To constrain the project's scope, initial tests will focus on academic papers—closed "systems" where terminology is highly structured and unambiguous.

Figure 1: User Login

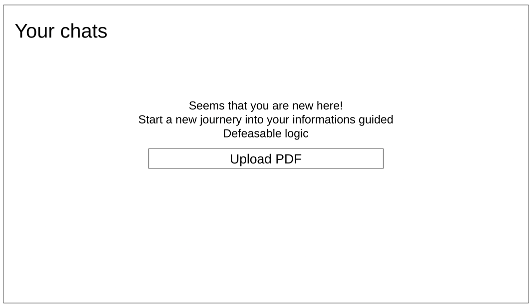

Figure 2: Upload a PDF

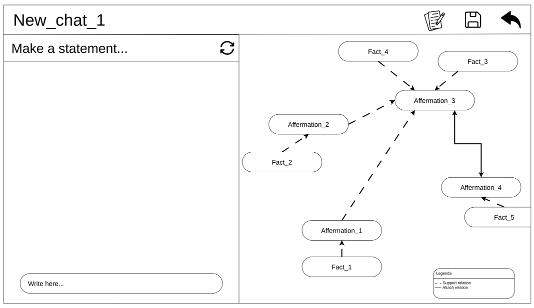

Figure 3: Dive into its system of **facts** and **argumentations**

2.2 Solutions and technologies *found so far*

NOTE

I am unsure to what extent I need to develop this project to achieve a strong grade. I would greatly appreciate your guidance in finding the right compromise between development effort and time constraints (see also Section 3). For this reason, the high-level presentation of the project ends before this box, and what follows are ideas to potentially enrich the scope of the work.

While a few days of dedicated exploration will be necessary, some preliminary research has already been conducted:

- **Structured and Extended Argumentation Frameworks:** To capture the nuances of defeasible reasoning, formalisms that build upon baseline abstract models are will explore. This includes **Structured Argumentation** frameworks that explicitly model defeasible logic, such as ASPIC+ or DeLP, as well as quantitative extensions to abstract frameworks, such as **Weighted Argumentation Frameworks** (WAF) or **Probabilistic Argumentation**. While many canonical engines are implemented in Java (e.g., *TweetyProject, Arg2P*), modern Python-based alternatives (e.g., *PyArg, DePYsible*) will be actively explored to streamline integration with the NLP extraction pipeline. Furthermore, for rapid prototyping and testing without developing immediately a custom human interface from scratch, tools offering out-of-the-box graph visualization—such as Python's *NetworkX* or the *TOAST* framework—will be utilized.
- **User interface:** To ensure the results are interpretable for everyone, a dedicated user interface is required, conceptualized as a basic web application. Drawing from recent personal projects, the frontend will likely be developed using React. This setup will exploit and study libraries such as React Flow to investigate how to map out and refine the reasoning results through a clear graphical representation.
- **Graph Database:** A lightweight graph database will be utilized to temporarily store the frameworks generated by the pipeline. While a dedicated database becomes crucial primarily for potential future expansions (see Section 2.3), it will immediately serve as a practical caching layer between user sessions.
- **Docker Compose:** To manage the distinct programming languages and technical scopes involved, the system will be deployed via a series of containers orchestrated by Docker Compose. This will streamline the development workflow and significantly enhance the reproducibility of the project.

About the testing part, benchmarks like **SciArg** (Scientific Argumentation), **UKP-Snopes / UKPLab Datasets**, **AbstRCT** and **CDCP** could be take in consideration as starting point.

2.3 Potential expansions

Building upon the general milestones outlined in Section 2.1, several upgrades can be introduced to enrich the project. Below are a few potential directions:

2.3.1 Handling multiple PDFs per session

Instead of restricting the knowledge base to a single PDF, this expansion would allow users to ingest a broader range of documents within a single session. Users could append documents after initializing a chat session and toggle specific sources on or off to observe how the resulting evaluation shifts.

The core challenge here lies in understanding how to join different nodes (such as facts, premises, and consequences) **extracted from separate PDFs**. Resolving this would be highly valuable, as it helps determine if—and how—distinct sources reinforce or contradict each other's claims, thereby providing a more comprehensive view of the arguments being established or defeated.

Translated to the web application, this feature would allow users to traverse directly between these affirmations, visualizing conflict points between opposing sources and tracking the precise chains of premises and consequences leading to a given conclusion. *This integration is expected to generate a vast network of nodes and arcs, especially with lengthy documents, which explains the conceptual origin of the project's name.*

In addition to this transition from local **systems** to a broader **galaxy**, another conceptual evolution could involve **expanding the types of source materials.** This means moving from texts with highly straightforward, structured terminology (e.g., scientific papers) to more complex contexts where meaning is less defined (e.g., general-use texts rich in semantic nuances).

2.3.2 Personal knowledge base

When a user proposes a statement, instead of merely generating a one-time support-rejection report, the resulting framework is stored in the database. This allows the creation of a personal knowledge base, enabling users to track how their reasoning evolves over time and, if desired, leverage this historical information during subsequent queries. **The challenge here lies in understanding how to implement this level of dynamism** within the software architecture.

2.3.3 Handling alternative formalisms

Different formalisms can yield conflicting structural outputs. Comparing different results given the same text input could be highly valuable to evaluate whether distinct formalisms converge on the same logical conclusions.

2.3.4 Integrating state-of-the-art NLP into the extraction pipeline

It is already apparent that one of the main challenges of this project lies in enabling the extraction pipeline to accurately capture the deep complexities inherent in natural language.

In the initial phases, traditional state-of-the-art (SOTA) NLP techniques prior to the advent of Transformers will be evaluated. This choice is motivated by their low computational requirements and, in the case of non-neural approaches, their highly interpretable and deterministic nature. However, because these methods can be limited in expressive power, the extraction pipeline may subsequently evolve to incorporate Transformer-based solutions. This transition could start with models like BERT to simplify the classification of terms within sentences—relying on deterministic rules to assemble the final argumentation framework—and potentially progress toward Small Language Models (SLMs).

Large Language Models (LLMs) are deliberately excluded from this development scope due to their massive computational demands, which conflict with the philosophy of building a lightweight tool accessible to everyone. Furthermore, generative Transformer models risk introducing hallucinations, which could compromise the factual integrity of the logical extraction process.

3 Project Timeline and Submission

My objective is **to complete this project by September 10, 2026, to be eligible for the Module 1 assessment and fully pass the course exam**. Given the scope of the project, I aim to finalize both the implementation and documentation by early September. I doubt I will be able to complete the entire workload by the end of July (due to concurrency of other exams), making an early September submission my primary target, assuming the impossibility of submission during August. I would highly appreciate your feedback on whether this proposed timeline and the project scope are compatible with your assessment criteria.